

End-of-Module-Exams 2023**Signature:** _____

I hereby confirm that all the pages of my exam paper
have been stapled to this cover sheet.

Number of supplementary answer sheets attached:

Total number of added pages:

Module:	w.BA.XX.3DA-WIN.XX / Data Analytics
Exam date / time:	25.01.2024 / 08:00 - 09:00
Exam duration:	60 Minutes
Total number of pages:	21 (incl. cover sheet)
Resources allowed:	Open book
Calculator:	Approved calculator according to 'Guidelines on Supplementary Materials'
Dictionary:	Yes

General information:

- Use a black or blue pen (indelible ink) to write your answers.
- Write your name on every page.
- Only write on the front side of each page.
- Supplementary answer sheets are provided by the invigilators. These must be clearly numbered and the number of supplementary answer sheets used must be noted on the cover sheet. Do not use your own sheets of paper.
- Sign the cover sheet before you hand in the exam.
- If you encounter an exam question that does not seem clear to you, you may feel the need to make assumptions. In such a case, summarize your assumptions in your answer.
- For remarks pertaining to specific questions, please refer to information included in the task.
- Toilet breaks are only allowed if the examination exceeds 90 minutes.
- Invigilators are not allowed to respond to any questions related to the content of the exam.
- If you wish to hand in your exam early, you may do so until 20 minutes before the official end of the examination.

Marked by:	Total points:	Grade:

Last Name:

First Name:

Answer Sheet – Please note that only the answers on this sheet will be assessed.

Grading Table

Performance Assessment	Weighting	Grade	Weighted Grade
Performance Assessment 1 Written Exam	70%		
Performance Assessment 2 Project Work (Group Work)	30%		
Performance Assessment 3 Submission of the solutions of at least 10 weekly exercises	Pass/Fail	Pass/Fail	-
		(Grade on cover sheet)	
Module grade (Evento)			⇌ Rounded ⇌

Marking Table

No.	Task	Points	Corrector's, initials
1	Single-Choice Questions		
Total Points			
Grade			

Overview

Task	Description	Points
1	Single-Choice Questions	60

TOTAL: 60 points

Last Name:

First Name:

Answer-Sheet (please mark the correct answer per question):

Question	a)	b)	c)	d)
Q1	☐	☐	☐	☐
Q2	☐	☐	☐	☐
Q3	☐	☐	☐	☐
Q4	☐	☐	☐	☐
Q5	☐	☐	☐	☐
Q6	☐	☐	☐	☐
Q7	☐	☐	☐	☐
Q8	☐	☐	☐	☐
Q9	☐	☐	☐	☐
Q10	☐	☐	☐	☐
Q11	☐	☐	☐	☐
Q12	☐	☐	☐	☐
Q13	☐	☐	☐	☐
Q14	☐	☐	☐	☐
Q15	☐	☐	☐	☐

Question	a)	b)	c)	d)
Q16	☐	☐	☐	☐
Q17	☐	☐	☐	☐
Q18	☐	☐	☐	☐
Q19	☐	☐	☐	☐
Q20	☐	☐	☐	☐
Q21	☐	☐	☐	☐
Q22	☐	☐	☐	☐
Q23	☐	☐	☐	☐
Q24	☐	☐	☐	☐
Q25	☐	☐	☐	☐
Q26	☐	☐	☐	☐
Q27	☐	☐	☐	☐
Q28	☐	☐	☐	☐
Q29	☐	☐	☐	☐
Q30	☐	☐	☐	☐

Question Section – Please note that this part will not be assessed.

General Information

- This exam consists of **single-answer questions only**.
- This means there is only one correct answer for each question.
- The possible answers for each question are listed as (a), (b), (c), and (d).
- If you provide more than one answer for a question, you will receive no marks for that question.
- If you do not answer a question, you will receive no marks for that question.
- Use the answer sheet for your solutions (**answer section** of the exam). This page belongs to the **question section** of the exam.

Overview

Task	Description	Points
1	Single-Answer Questions	60

TOTAL: 60 points

Single-Answer Questions

(60 points)

Categorization of Data & Data Collection Methods								
Q1: Consider the following table containing smartphone data. Which of the statements with regard to the “scale of measure” of the variables is <u>not</u> correct?								2 points
	Brand	Storage	RAM	Display_Size	OS	Price_Category	Price	
0	Apple	256	6	6.1	iOS	High	999	
1	Samsung	128	8	6.4	Android	High	899	
2	OnePlus	256	12	6.7	Android	High	799	
3	Xiaomi	128	6	6.5	Android	Mid	599	
4	Google	64	8	5.8	Android	Mid	699	
a) The variable <i>Brand</i> has unordered category labels and is, therefore, nominal scaled. b) The variable <i>Price</i> has an absolute (or true) zero and is, therefore, ratio scaled. c) The variable <i>OS</i> has ordered category labels and is, therefore, interval scaled. d) The variable <i>Price Category</i> has ordered category labels and is, therefore, ordinal scaled.								
Q2: The text item “iPhone 13 Pro” must be extracted from the following HTML file using an XPath. Which is the correct XPath for extracting this text item?								2 points
<pre><!DOCTYPE html> <html> <head> <title>Sample HTML</title> </head> <body> <div id="content"> <h1>welcome to Our website</h1> <p>Explore our latest products and services.</p> <li class="product">iPhone 13 Pro <li class="product">Samsung Galaxy S21 Ultra <li class="product">Google Pixel 6 Pro <div class="footer"> <p>For any inquiries, contact us at support@ourwebsite.com</p> </div> </div> </body> </html></pre>								
a) /html/body/div/ul/li[1]/text() b) /html/head/body/div/div/ul/li[1]/text() c) /html/body/div/ul/li[2]/text() d) /html/head/body/div/div/ul/li[2]/text()								

Q3: Which statement with regard to the JSON file below (and the JSON file format in general) is not correct?

2 points

```
{  
  "name": "John"  
  "age": 30  
  "city": "New York"  
  "hobbies": ["Reading", "Traveling", "Cooking"]  
  "street": "123 Main St"  
  "email": "john@example.com"  
  "phone": "123-456-7890"  
}
```

- a) In this JSON file, data is available as 'name/value' (also called 'key/value') pairs.
- b) This JSON file has an invalid structure due to missing commas.
- c) In a JSON file, square brackets are used to hold an array (a list of values).
- d) JSON files can only represent data in a single line without any nested structures.

Q4: Consider the following table, which contains a five-point Likert scale from an online survey. Which statement is not correct with regard to the scale of measure of such Likert-scale data?

2 points

Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1	2	3	4	5

- a) Likert-scale data have ordered category labels.
- b) Analyzing Likert-scale data requires specific statistical methods (e.g. correlation based on rank sum, ordinal regression).
- c) Likert-scale data are numerical data.
- d) The distance between levels (e.g., between *Disagree* and *Neutral* or between *Neutral* and *Agree*) is not uniform or quantifiable and, therefore, not meaningful.

Data Types, Data Structures, & File Formats in Python	
Q5: Which data types are provided by the print() function in the code below? <pre>a = 24 b = 100 / 15 * 2 c = 'dog' d = False e = 7 == 8 f = 6 > 5 g = int(str(71)) print(type(a), type(b), type(c), type(d), type(e), type(f), type(g))</pre> a) int, float, str, False, False, True, int b) int, float, str, bool, bool, bool, str c) int, int, str, False, False, True, int d) int, float, str, bool, bool, bool, int	2 points
Q6: What type will the following expression return? <pre>obj = [{'pets': ['cat', 'dog', 'bird']}, [1,2,3,4], ('A', 'B', 'C'), {'mouse', 'turtle', 'fish'}] print(type(obj[1]))</pre> a) <class 'tuple'> b) <class 'list'> c) <class 'set'> d) <class 'dict'>	2 points

Q7: What is the output of the following Python code? Note that <code>.get()</code> is used here to get the values of the key stored in 'key_sel'. Also note that <code>list()</code> converts an object to a list, for example, <code>list({1,2,3,4})</code> returns <code>[1, 2, 3, 4]</code>. <pre>import pandas as pd obj = { 'id': [1, 2, 3, 4], 'make': ['Toyota', 'BMW', 'Ford', 'Audi'], 'model': ['Camry', 'X5', 'Focus', 'A4'], 'year': [2018, 2020, 2019, 2017], 'color': ['Red', 'Black', 'Blue', 'Silver'] } cars = pd.DataFrame(obj) key_sel = list(cars.keys())[2] val_sel = list(cars.get(key_sel)) print(val_sel)</pre> a) ['Camry', 'X5', 'Focus', 'A4'] b) ['Toyota', 'BMW', 'Ford', 'Audi'] c) [2018, 2020, 2019, 2017] d) ['id', 'make', 'model', 'year', 'color']	2 points
Q8: Imagine you must read a file named 'myfile' to a pandas dataframe. The pandas library has already been imported as 'pd'. There is no file extension. You opened the file with a text editor and looked at its structure (see below). Which line of code reads the data correctly? <pre># Structure of the file named 'myfile': { "name": "Alice", "age": 30, "city": "New York", "interests": ["reading", "traveling", "cooking"] }</pre> a) <code>data = pd.read_csv('myfile', encoding='utf-8')</code> b) <code>data = pd.read_xml('myfile', encoding='utf-8')</code> c) <code>data = pd.read_html('myfile', encoding='utf-8')</code> d) <code>data = pd.read_json('myfile', encoding='utf-8')</code>	2 points

Data Preparation & Combining & Organizing Data	
Q9: Consider the following Python code for creating and combining the two data frames named 'df1' and 'df2'. How many rows does the output data frame 'df3' contain, and how many columns? Note that 'number of columns' means without the index (i.e., 0,1,2,3,4). <pre>import pandas as pd # 1st DataFrame df1 = pd.DataFrame({ 'Fruit': ['Apple', 'Banana', 'Orange', 'Grape'], 'Color': ['Red', 'Yellow', 'Orange', 'Purple'], 'Taste': ['Sweet', 'Sweet', 'Citrus', 'Sweet']}) # 2nd DataFrame df2 = pd.DataFrame({ 'Fruit': ['Kiwi', 'Pineapple', 'Strawberry', 'Mango'], 'Color': ['Brown', 'Yellow', 'Red', 'Orange'], 'Taste': ['Tart', 'Sweet', 'Sweet', 'Sweet']}) # Combining the DataFrames df3 = pd.concat([df1, df2], ignore_index=True) print(df3)</pre> a) The data frame 'df3' has 7 rows and 5 columns. b) The data frame 'df3' has 5 rows and 5 columns. c) The data frame 'df3' has 8 rows and 3 columns. d) The data frame 'df3' has 4 rows and 6 columns.	2 points
Q10: Consider the pandas dataframe below. Which values are shown by the print() function? Note that the values are rounded to one decimal place. <pre>import pandas as pd data = { 'co2': [95, 90, 99, 104, 105, 94, 99, 104], 'model': ['Citigo', 'Fabia', 'Fiesta', 'Rapid', 'Focus', 'Mondeo', 'Octavia', 'B-Max'], 'make': ['Skoda', 'Skoda', 'Ford', 'Skoda', 'Ford', 'Ford', 'Skoda', 'Ford']} df = pd.DataFrame(data) print(df[['make', 'co2']].groupby(['make']).mean().round(1))</pre> a) Ford 98.4 Skoda 96.2 b) Ford 102.4 Skoda 98.7 c) Ford 100.5 Skoda 97.0 d) Ford 99.6 Skoda 97.2	2 points

Q11: Consider the pandas dataframe below (name = 'data') and the Python code for the selection. Which pets are displayed by the print() function in the Python code? Remember that the `|` operator inside the `.loc` method means 'OR' and the `!=` operator means 'NOT EQUAL'.

2 points

	name	type	age	breed
0	Buddy	Dog	5	NaN
1	Misty	Cat	3	Siamese
2	Max	Dog	7	German Shepherd
3	Charlie	Rabbit	2	NaN
4	Lucy	Cat	4	Siamese
5	Hasso	Dog	3	Labrador

```
subset = data.dropna().loc[(data['age'] >= 3) | (data['type'] != 'Rabbit')]  
print(list(subset['name']))
```

- a) ['Misty', 'Max', 'Lucy', 'Hasso']
- b) ['Buddy', 'Misty', 'Max', 'Charlie']
- c) ['Max', 'Charlie', 'Lucy', 'Hasso']
- d) ['Buddy', 'Max', 'Charlie', 'Lucy']

Q12: What is the output of the print() function in the following Python code?

2 points

```
import pandas as pd
import numpy as np

# Creating a DataFrame for car data
data = {
    'Brand': ['Toyota', 'BMW', 'Toyota', 'BMW', 'Toyota', 'BMW'],
    'Model': ['Camry', 'X5', 'Corolla', 'X3', 'Corolla', 'X5'],
    'Price': [35000, 45000, 40000, 48000, 42000, 54000],
    'Sales': [500, 300, 450, 200, 600, 250]
}
cars = pd.DataFrame(data)

# Creating a pivot table to summarize the values
pivot_cars = pd.pivot_table(cars,
                             index=['Brand', 'Model'],
                             values=['Price', 'Sales'],
                             aggfunc=np.mean)

print(pivot_cars)
```

a)

		Price	Sales
Brand	Model		
BMW	X3	48000	200
	X5	99000	550
Toyota	Camry	35000	500
	Corolla	82000	1050

b)

		Price	Sales
Brand	Model		
BMW	X3	48000	200
	X5	45000	300
Toyota	Camry	35000	500
	Corolla	82000	1050
VW	Golf	54000	250

c)

		Price	Sales
Brand	Model		
BMW	X3	48000	200
	X5	49500	275
Toyota	Camry	35000	500
	Corolla	41000	525

d)

		Price	Sales
Brand	Model		
BMW	X3	48000	200
	X5	45000	300
Toyota	Camry	35000	500
	Corolla	41000	525
VW	Golf	54000	250

Exploratory Data Analysis (EDA)

Q13: Consider the following contingency table. It contains data from an online survey. Which statement with regard to this contingency table is not correct?

2 points

Observed frequencies

	Boy	Girl	sum
like Snickers	86	60	146
doesn't like Snickers	16	38	54
sum	102	98	200

- a) 43% of all respondents are boys who like Snickers.
- b) 27% of all respondents don't like Snickers.
- c) 30% of all respondents are girls who like Snickers.
- d) 16% of all respondents are boys who don't like Snickers.

Q14: Consider the following data representing a sample of house prices in Switzerland. Which is the correct value of the sample standard deviation of house prices in the sample? Note that the values in the answer options are rounded to one decimal place.

2 points

House_Prices

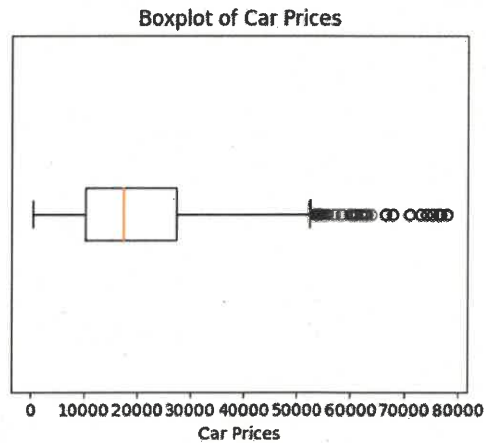
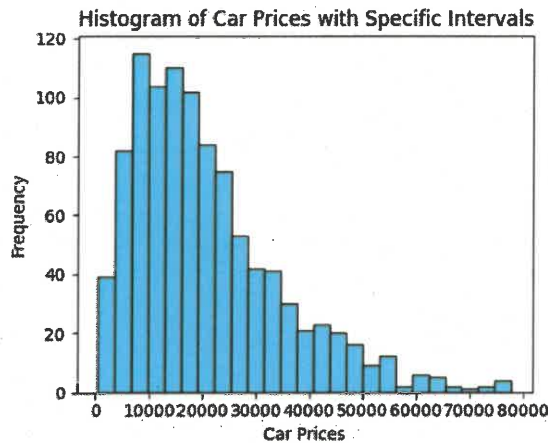
0	1500000
1	1600000
2	1700000
3	1800000
4	1900000

- a) 172465.2
- b) 158113.9
- c) 164257.8
- d) 170854.2

Q15: Consider the following graphics showing the distribution of car prices (in U.S. dollars) and the measures of central tendency of the price distribution. Which statement with regard to these graphics and measures is not correct?

2 points

25th percentile (1st Quartile): 10262
50th percentile (Median): 17291
75th percentile (3rd Quartile): 27422



- a) The car prices show a right-skewed distribution.
- b) The interquartile range (IQR) is \$18160.
- c) 50% of the prices are in the range between \$10262 and \$27422.
- d) From the graphics, it can be assumed that the mean car price is > \$17291.

Q16: Consider the following correlation heatmap based on numerical environmental data and the description of variables. Which interpretation of the correlation heatmap is not correct? Note that all correlation coefficients (r) provided in the answer options were taken from the correlation heatmap and can be considered correct. The statistical significance of the correlations must not be considered here.

2 points

Description of variables:

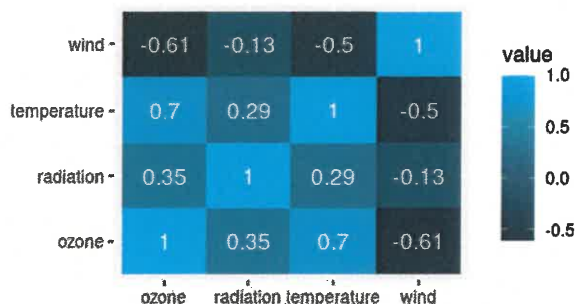
wind = wind speed in km/h

temperature = temperature in degrees Celsius

radiation = solar radiation in kWh/m²

ozone = ozone concentration in g/m³

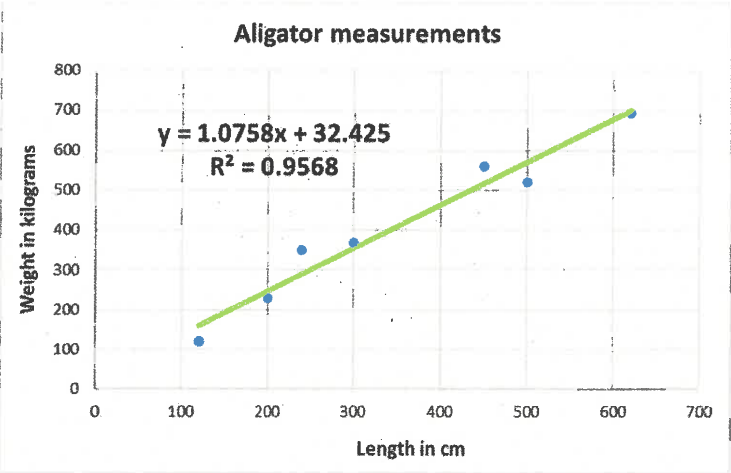
Correlation heatmap



- a) The variable "radiation" is positively correlated with the variable "ozone" ($r=0.35$), indicating that higher solar radiation correspond to higher ozone concentration.
- b) The variable "wind" is negatively correlated with the variable "temperature" ($r=-0.5$), suggesting that higher wind speeds are associated with lower temperatures.
- c) The variable "temperature" shows a positive relationship with the variable "ozone" ($r=0.7$), indicating that higher temperatures correspond to lower ozone concentrations.
- d) The variable "radiation" shows a negative relationship with the variable "wind" ($r=-0.13$), suggesting that higher wind speed is associated with lower solar radiation.

Representation & Analysis of Geographical Data	
Q17: Which statement with regard to the GIS data format below is <u>not</u> correct? Polygon  POLYGON ((30 10, 40 40, 20 40, 10 20, 30 10))  POLYGON ((35 10, 45 45, 15 40, 10 20, 35 10), (20 30, 35 35, 30 20, 20 30)) a) This GIS data format is referred to as GeoJSON. b) This data format is used to store vector data. c) The coordinates, e.g., 30 10, 40 40, 20 40 refer to points that shape the polygon. d) This data format can also be used to store point and line geometry types.	2 points
Q18: Consider the following Python code and the parts of the code denoted Part (1) ... Part (4). Which statement below with regard to this Python code is <u>not</u> correct? <pre>import json import urllib import requests # Part (1) base_url= "https://api3.geo.admin.ch/rest/services/api/SearchServer?" # Part (2) parameters = {"searchText": "Marktstrasse 14, 8500 Frauenfeld", "origins": "address", "type": "locations", } # Part (3) r = requests.get(f"{base_url}{urllib.parse.urlencode(parameters)}") # Part (4) data = json.loads(r.content) df = pd.DataFrame.from_dict(list(data.values())[0][0], orient='columns') df.iloc[[1,4,5,6,11,12],:1]</pre> a) The Python code is used to geocode a single address using the GeoAdmin Web API. b) The focus of the GeoAdmin Web API is to offer geospatial data, maps, and services for different European countries. c) The requests.get() method in Part (3) sends a GET request to the constructed URL. d) In Part (4), df.iloc[[1,4,5,6,11,12],:1] is used to select relevant data from the server response stored in the data frame named 'df'.	2 points

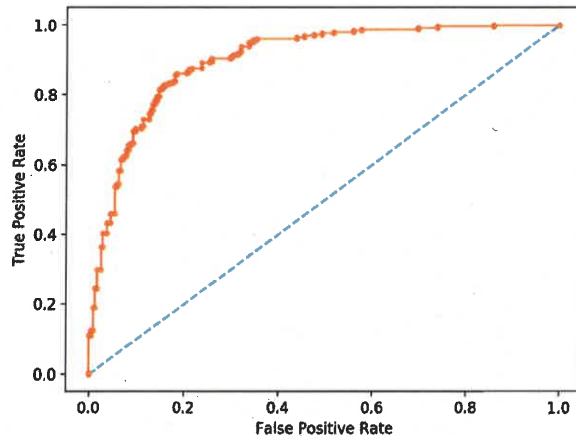
Statistical Hypothesis Testing																		
Q19: Which statement is <u>not</u> correct regarding an unpaired t-test? a) An unpaired t-test compares the means of two independent groups. b) It assesses whether the means of two samples are statistically different from each other. c) It is used when the individuals in one group are unrelated to those in the other group. d) A before/after study, such as a cattle feeding trial, is an example of an unpaired t-test.		2 points																
Q20: Consider the following contingency table of observed frequencies from a survey of people's beverage preferences. For a Chi-squared test, the expected frequencies must be calculated. Which are the correct expected frequencies? Note that the values in the answer options are rounded to two decimal places. Observed frequencies <table><tr><td></td><td>wine</td><td>beer</td><td>sum</td></tr><tr><td>men</td><td>64</td><td>78</td><td>142</td></tr><tr><td>women</td><td>75</td><td>62</td><td>137</td></tr><tr><td>sum</td><td>139</td><td>140</td><td>279</td></tr></table> a) Expected frequencies: [[71.75 72.25] [67.25 69.75]] b) Expected frequencies: [[72.75 74.25] [64.25 66.75]] c) Expected frequencies: [[70.75 71.25] [68.25 68.75]] d) Expected frequencies: [[69.75 70.25] [65.25 72.75]]			wine	beer	sum	men	64	78	142	women	75	62	137	sum	139	140	279	2 points
	wine	beer	sum															
men	64	78	142															
women	75	62	137															
sum	139	140	279															

Regression Analysis	
Q21: Consider the following linear regression, displaying the relationship between the weight (target variable = y) and length (explanatory variable = x) of randomly selected alligators in a wildlife sanctuary. The equation and R^2 of the regression are presented in the graphic below. When the regression model is used for predicting the weights of alligators, which statement is <u>not</u> correct? Note that the predicted weights in the answer options are rounded without decimal places.  a) An alligator with a length of 550 cm has a predicted weight of 624 kilograms. b) An alligator with a length of 650 cm has a predicted weight of 732 kilograms. c) An alligator with a length of 450 cm has a predicted weight of 589 kilograms. d) An alligator with a length of 340 cm has a predicted weight of 398 kilograms.	2 points
Q22: Given the regression equation for car prices based on mileage shown below, what does this equation imply? $y = -0.22x + 45000$ where: y = estimated car price (in \$) x = mileage (in kilometers) a) For every decrease of 0.22 units in mileage, the car price decreases by \$45. b) For every increase of 1 unit in mileage, the car price decreases by \$0.22. c) For every increase of \$0.22 in car price, the mileage decreases by 4.5 units. d) For every increase of 1 unit in mileage, the car price increases by \$0.22.	2 points

Q23: Imagine you have two numerical variables. You perform a correlation analysis and a regression analysis based on these two variables. If the coefficient of determination (R^2) equals 1, then the correlation coefficient (r) ... a) ... must also be equal to 1. b) ... can be either -1 or +1. c) ... can be any value between -1 and +1. d) ... must be -1.	2 points									
Classification Analysis										
Q24: Consider the following confusion matrix of a classification model in which TN = True Negative, FN = False Negative, FP = False Positive, and TP = True Positive. Which is the correct value of the 'Accuracy' (rounded to two decimal places)? <table><tr><td></td><td>Predicted: Not Spam</td><td>Predicted: Spam</td></tr><tr><td>Actual: Not Spam</td><td>850 (TN)</td><td>50 (FP)</td></tr><tr><td>Actual: Spam</td><td>30 (FN)</td><td>950 (TP)</td></tr></table> a) 0.96 b) 0.92 c) 0.89 d) 0.94		Predicted: Not Spam	Predicted: Spam	Actual: Not Spam	850 (TN)	50 (FP)	Actual: Spam	30 (FN)	950 (TP)	2 points
	Predicted: Not Spam	Predicted: Spam								
Actual: Not Spam	850 (TN)	50 (FP)								
Actual: Spam	30 (FN)	950 (TP)								
Q25: You can generate hundreds of trees in a single Random Forest model (say T_1, $T_2 \dots T_n$). Which of the following four statements about an individual (T_k) tree in a Random Forest model are true? Remember that the term <i>features</i> used in the context of machine learning (ML) is a synonym for the term <i>variables</i> used in statistics. Statements: (1) "An individual tree is built on a randomly selected subset of features." (2) "An individual tree is built on all features." (3) "An individual tree is built on a randomly selected subset of observations." (4) "An individual tree is built on the full set of observations." a) Statements (1) and (3) are true. b) Statements (1) and (4) are true. c) Statements (2) and (3) are true. d) Statements (2) and (4) are true.	2 points									

Q26: Consider the following figure showing a ROC curve from a classification model. Which statement about the ROC curve and related measures, such as the area under the ROC curve (AUC), is not correct?

2 points



- a) The area under the ROC curve (AUC) represents the model's ability to discriminate between classes, with a higher AUC indicating better model performance.
- b) A ROC curve shows the model performance of a classification model for different classification thresholds.
- c) A classification model with purely random predictions is expected to provide a ROC curve that lies on the diagonal line.
- d) Classification models that provide ROC curves closer to the top-left corner of the ROC plot indicate a weaker model performance compared to other classification models.

Cluster Analysis

Q27: How is the optimal number of clusters typically determined in k-means clustering?

2 points

- a) By selecting the number of clusters that maximizes the within-cluster variance.
- b) By employing, for example, an elbow plot.
- c) By using cross-validation with an 80% sample for training and a 20% sample for testing.
- d) By selecting the number of clusters that maximizes the coefficient of determination (R^2).

Q28: Consider the following Figure 1 showing a silhouette plot and the output of silhouette analyses based on different numbers of clusters (from $k=3 \dots k=6$). Figure 1 represents one of the k-means clustering results shown in Figure 2. Which interpretation of silhouette plots in general and the results shown in Figure 1 and Figure 2, in particular, is not correct?

2 points

Figure 1: Silhouette Plot

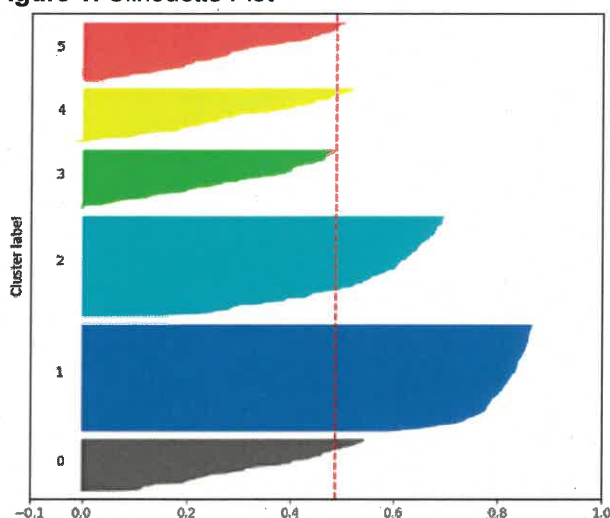


Figure 2: Silhouette Scores for k-means algorithms with $k=3 \dots k=6$:

For `n_clusters = 3` The average silhouette_score is : 0.5882
For `n_clusters = 4` The average silhouette_score is : 0.6505
For `n_clusters = 5` The average silhouette_score is : 0.5614
For `n_clusters = 6` The average silhouette_score is : 0.4857

- a) Figure 1 shows six clusters along with their observations and silhouette coefficient values. The silhouette score is shown as a red dotted line in this figure.
- b) Silhouette coefficients near +1 indicate that the sample is far away from the neighboring clusters, while a coefficient of 0 indicates that the sample is on or very close to the decision boundary between two neighboring clusters.
- c) In Figure 1, the clusters labeled '1' and '2' are the largest clusters as they have the largest number of observations compared to the other clusters.
- d) The k-means clustering algorithm with $k=6$ is the best choice because it provides the lowest silhouette score, indicating that the clusters are better separated than the k-means clustering algorithms with $k=3$, $k=4$, and $k=5$.

Text Mining & Sentiment Analysis	
Q29: Consider the sentence, "Paul wanted a better job.". Which of the following options corresponds to complete preprocessing of the sentence according to the steps for text preprocessing presented in class? a) ["paul", "want", "good", "job"] b) ["paul", "want", "a", "good", "job"] c) ["Paul wanted a better job." d) ["Paul", "wanted", "a", "better", "job"]	2 points
Q30: Using the bag-of-words (BoW) algorithm, what is the correct numerical representation of the sentence, "the cat chased the cat around the cat"? (Hint: Ignore any preprocessing and consider the sentence as the basis for the vocabulary. Represent the sentence in the order of the words as they occur, i.e., no alphabetical order of words is required for the BoW representation.). a) [2, 3, 1, 1] b) [3, 2, 1, 2] c) [3, 3, 1, 1] d) [2, 2, 2, 1]	2 points

END OF EXAM